Time series: analysis of real-life data

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MA3508, MA4508, MAT508
Lecture notes 14

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The homework is provided on the final slides

Analysis of real-life time series

- **Plot the given time series** x_t and guess a model. Usually, x_t is the sum of a trend, seasonal or cyclical components and an irregular or noise component, see Lecture 1.
- If x_t is non-"stationary", then **find a transformation to obtain a "stationary" series**. If the variance $\mathrm{Var}(x_t)$ is increasing and $x_t > 0$, take the logarithm $y_t = \log(x_t)$. If the mean function $\mathbf{E}x_t$ is not constant, take the differences $y_t = x_t - x_{t-1}$.
- **Compute the correlogram** for y_t and guess a model. If it contains slightly damped oscillations, then y_t has a significant cyclical component. If it is slowly decreasing, then an AR(p) model may be relevant. If it is quickly decreasing, then an MA(q) model may be relevant.
- **Compute the periodogram** for y_t and guess a model. Identify frequencies which have a large contribution.

Airpass data

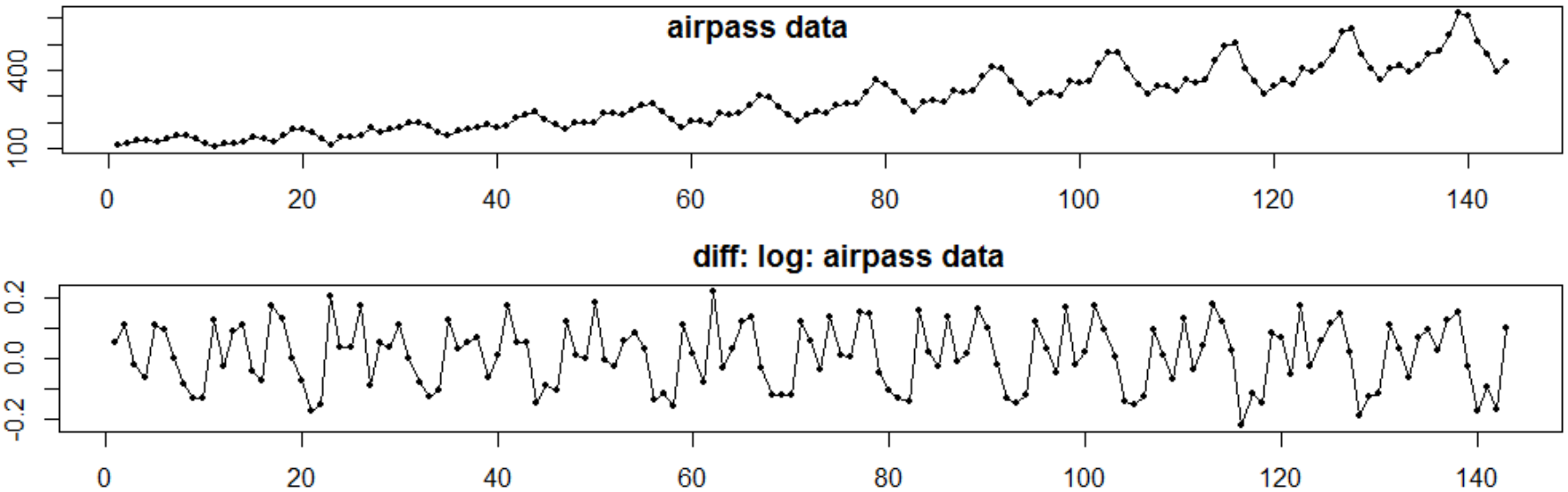
Consider the time series of monthly airline passenger numbers, 1949-1960 (Box & Jenkins airline data).

Taking the logarithm and then taking differences, we obtain a stationary time series.

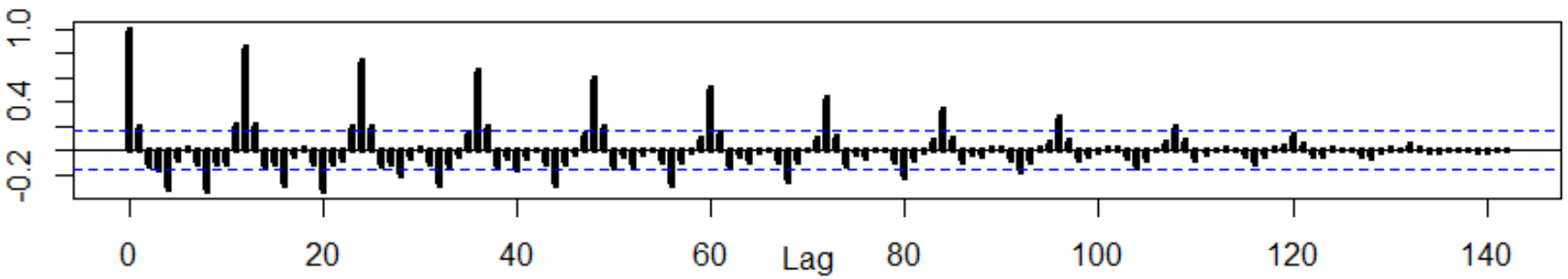
®

```
airpass=read.csv("http://ssa.cf.ac.uk/ma0367/airpass.csv",
sep=";",header=TRUE); print(airpass)
plot(airpass[,3],type="l", main="airpass data")
points(airpass[,3],pch=19,cex=0.6)
dev.new()
plot(log(airpass[,3]),type="l", main="log: airpass data")
points(log(airpass[,3]),pch=19,cex=0.6)
dev.new(); y=diff(log(airpass[,3]))
plot(y,type="l", main="diff: log: airpass data")
points(y,pch=19,cex=0.6)
```

Airpass data: correlogram



® *corrgram=acf(y,length(y),lwd=4)*



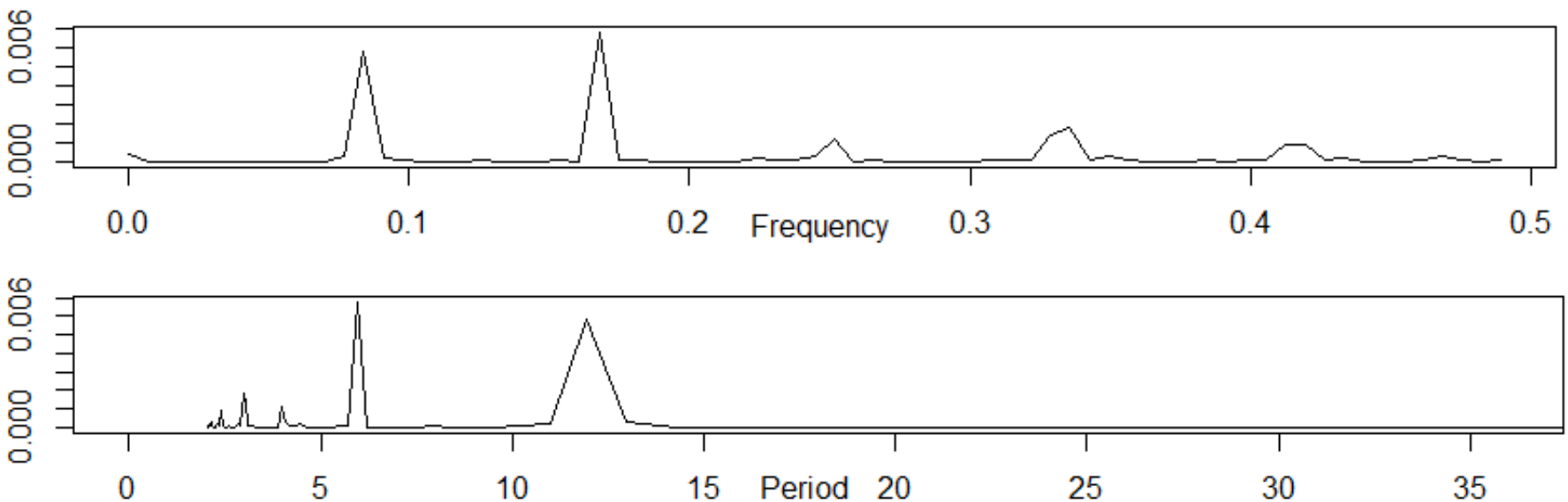
Relevant models: $y_t = a_{12}y_{t-12} + \epsilon_t$, or

$y_t = a_1 y_{t-1} + a_{12}y_{t-12} + \epsilon_t$, or

$y_t = a_1 y_{t-1} + a_{12}y_{t-12} + a_{13}y_{t-13} + \epsilon_t$

Airpass data: periodogram

```
® Pgram=function(xts,title="time series")
{   n=length(xts); I = abs(fft(xts-mean(xts)))**2/n;
  P = (4/n)*I[1:(n/2)]; nu = c(0:(n/2-1))/n;
  plot(nu, P, type="l", xlab="Frequency",main=title,
     ylab="Scaled Periodogram"); return( cbind(nu,P) )
}
pgram=Pgram(y,"Periodogram of series y"); dev.new();
plot(1/pgram[,1], pgram[,2], type="l", xlab="Period" ,
     ylab="Scaled Periodogram",xlim=c(0,36))
```



Airpass data: estimation

® *arima(y,order=c(1,0,0))*
arima(y,order=c(2,0,0))
arima(y,order=c(12,0,0))
arima(y,order=c(1,0,0),seasonal=list(order=c(1,0,0),period=12))
arima(y,order=c(1,0,1),seasonal=list(order=c(1,0,0),period=12))
arima(y,order=c(1,0,0),seasonal=list(order=c(2,0,0),period=12))

Estimated models

$$y_t \overset{(1,0,0)}{=} 0.20y_{t-1} + \epsilon_t,\ \hat{\sigma}^2 = 0.0108$$

$$y_t \overset{(2,0,0)}{=} 0.24y_{t-1} - 0.17y_{t-2} + \epsilon_t,\ \hat{\sigma}^2 = 0.0105$$

$$y_t \overset{(12,0,0)}{=} -0.22y_{t-1} - 0.28y_{t-2} - \ldots - 0.20y_{t-11} + 0.60y_{t-12} + \epsilon_t,\ \hat{\sigma}^2 = 0.0016$$

$$(1 + 0.29B)(1 - 0.93B^{12})y_t \overset{(1,0,0)(1,0,0)}{=} \epsilon_t,\ \hat{\sigma}^2 = 0.0018$$

$$(1 - 0.74B)(1 - 0.91B^{12})y_t \overset{(1,0,1)(1,0,0)}{=} \epsilon_t - 1.0\epsilon_{t-1},\ \hat{\sigma}^2 = 0.0017$$

Airpass data: Model identification

A good model has ...

- a small number of parameters, N (for easier interpretation)
- a small noise variance, $\hat{\sigma}^2$
- a large log likelihood, LL
- a small Akaike Information Criterion, AIC

More about AIC, see `https://www.otexts.org/fpp/8/6`

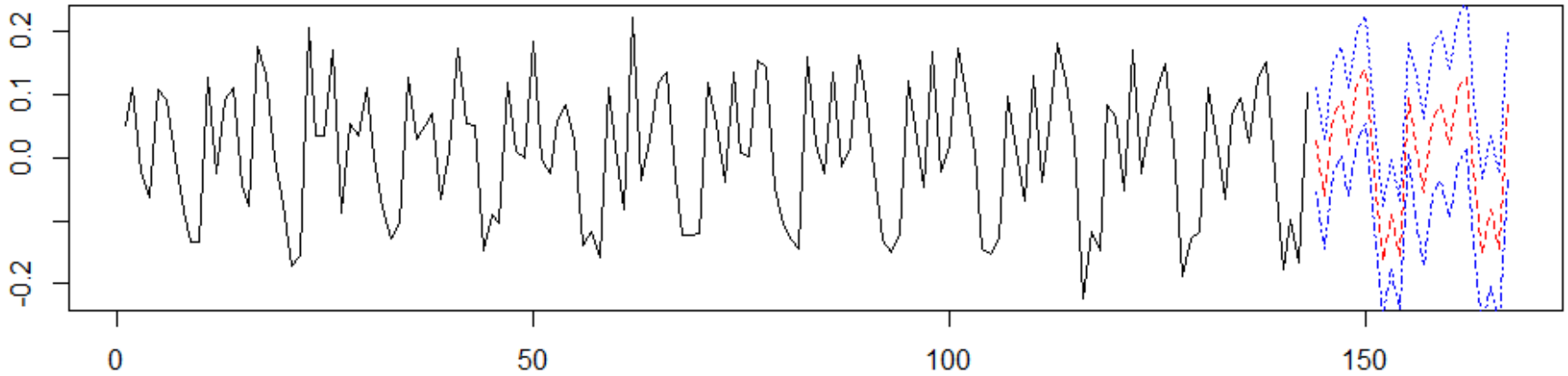
model	N	$\hat{\sigma}^2$	LL	AIC
ARMA(1,0)	1	0.01082	120.69	-235.39
ARMA(2,0)	2	0.01050	122.80	-237.60
ARMA(12,0)	12	0.00164	246.66	-465.31
SARIMA(1,0,0)(1,0,0)^{12}	2	0.00177	237.98	-467.96
SARIMA(1,0,1)(1,0,0)^{12}	3	0.00171	242.45	-474.90

Airpass data: forecasting

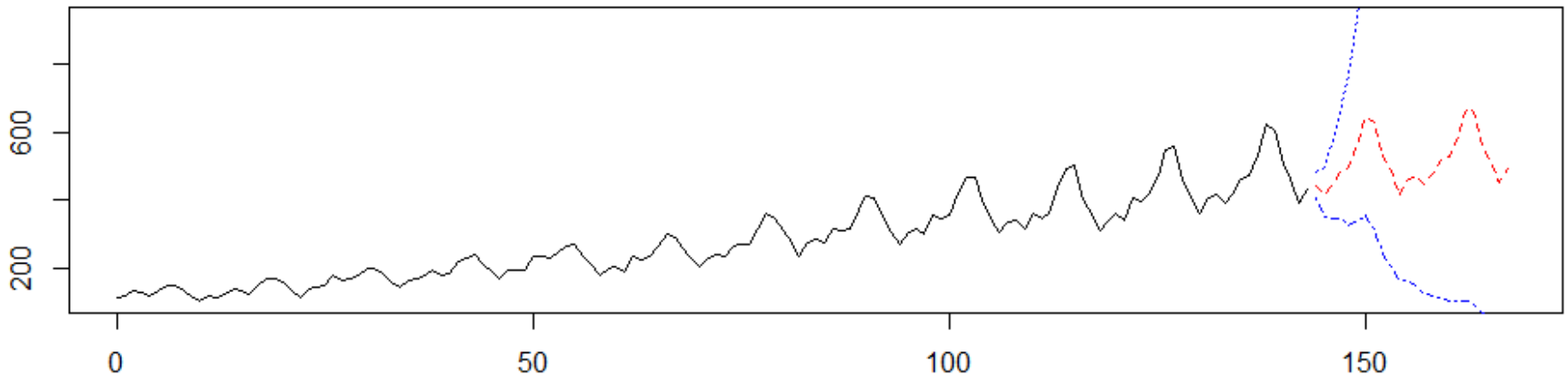
```
® n=length(y); h=24;
model=arima(y,order=c(1,0,0),seasonal=list(order=c(1,0,0),period=12))
pred=predict(model, n.ahead=h)$pred
s.pred=predict(model, n.ahead=h, se.fit=T)$se
plot(1:n, y, type="l", xlim=c(1,n+h),main="diff log airpass")
lines(n+(1:h), pred, type="l", lty=2, col="red")
lines(n+(1:h), pred+1.96*s.pred, type="l", lty=3, col="blue")
lines(n+(1:h), pred-1.96*s.pred, type="l", lty=4, col="blue")
dev.new()
plot(0:n, airpass[,3], type="l", xlim=c(1,n+h),
   ylim=range(airpass[,3])*c(1,1.5),main="airpass")
lines(n+(1:h), exp(log(airpass[1+n,3])
   +cumsum(pred)), type="l", lty=2, col="red")
lines(n+(1:h), exp(log(airpass[1+n,3])
   +cumsum(pred+1.96*s.pred)), type="l", lty=3, col="blue")
lines(n+(1:h), exp(log(airpass[1+n,3])
   +cumsum(pred-1.96*s.pred)), type="l", lty=4, col="blue")
```

Airpass data: forecasting graph

Forecasting of diff-log-airpass data, SARIMA(1,0,0)(1,0,0)



Forecasting of airpass data, log-SARIMA(1,1,0)(1,0,0)



More sophisticated analysis `http://www.uio.no/studier/emner/matnat/math/STK4060/v06/R%20code%20for%20exercise%206.7-8.txt`

Global Earth temperature

Consider the Earth temperature records from the website `http://vortex.nsstc.uah.edu/` (National Space Science and Technology Center, NASA, USA).

The records represent the temperature on Earth and some of its parts since December 1978.

The series consist of the so-called temperature anomalies rather than the absolute temperatures (temperature anomalies are computed relative to the base period 1951-1980).

Parts of Earth are Earth land, Earth ocean, Tropics, North Pole, South Pole, Northern Hemisphere, Southern Hemisphere.

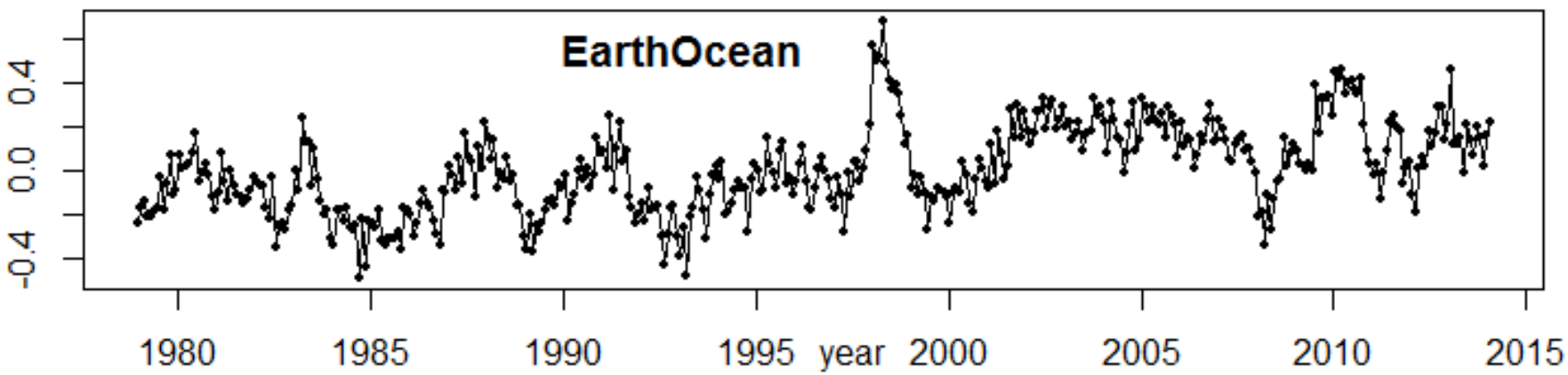
Datafile:

`https://www.nsstc.uah.edu/data/msu/v6.0/tlt/uahncdc_lt_6.0.txt`

Earth Ocean temperature: graph

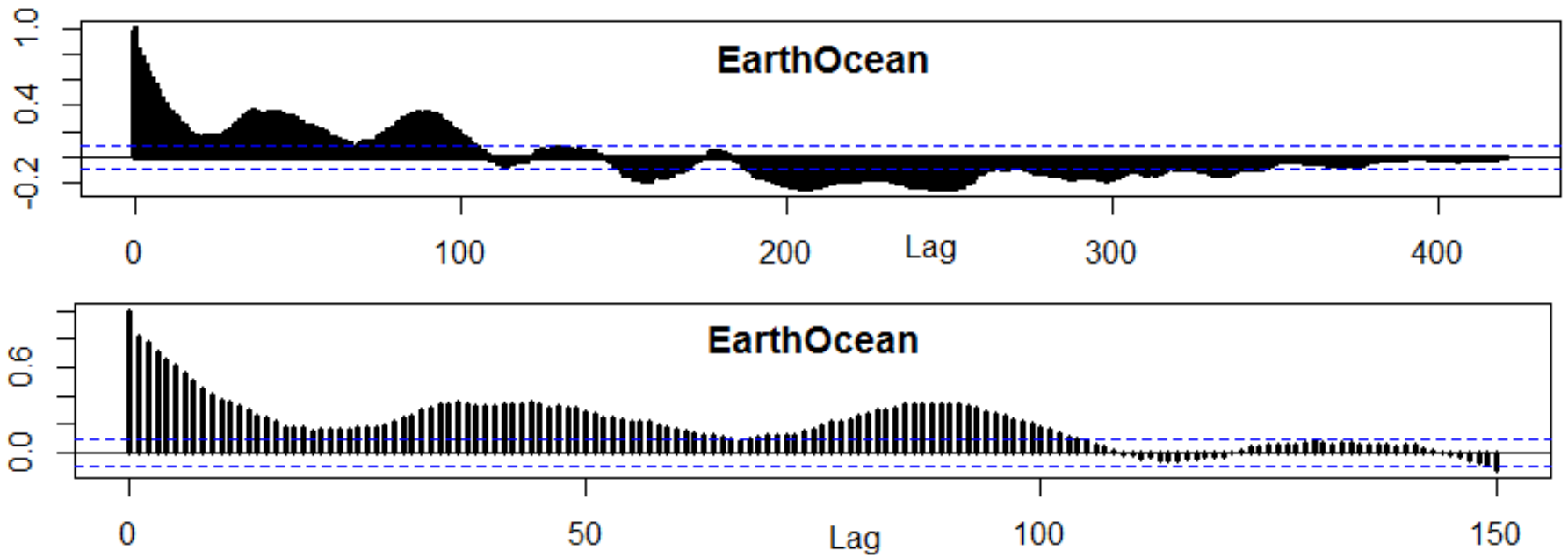
®

```
data=read.table("http://ssa.cf.ac.uk/ma0367/globtemp.txt",
   header=TRUE)
print(data[1:4,]); colindex=5;
x=data[,colindex]; xname=names(data)[colindex];
n=length(x);  year=data[,1]; month=data[,2];
t=year+month/12-1/24
plot(t,x,type="l",main=xname,xlab="year")
points(t,x,pch=19,cex=0.6)
```

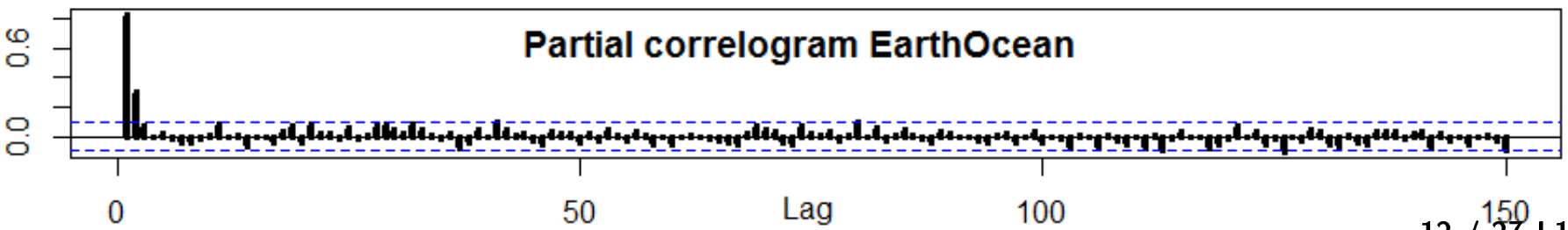


Earth Ocean temperature: correlogram

® *corrgram=acf(x,n,lwd=4,main=xname); dev.new();*
corrgram150=acf(x,150,lwd=4,main=xname); dev.new();
pacf(x,150,lwd=4,main=paste("Partial correlogram",xname))

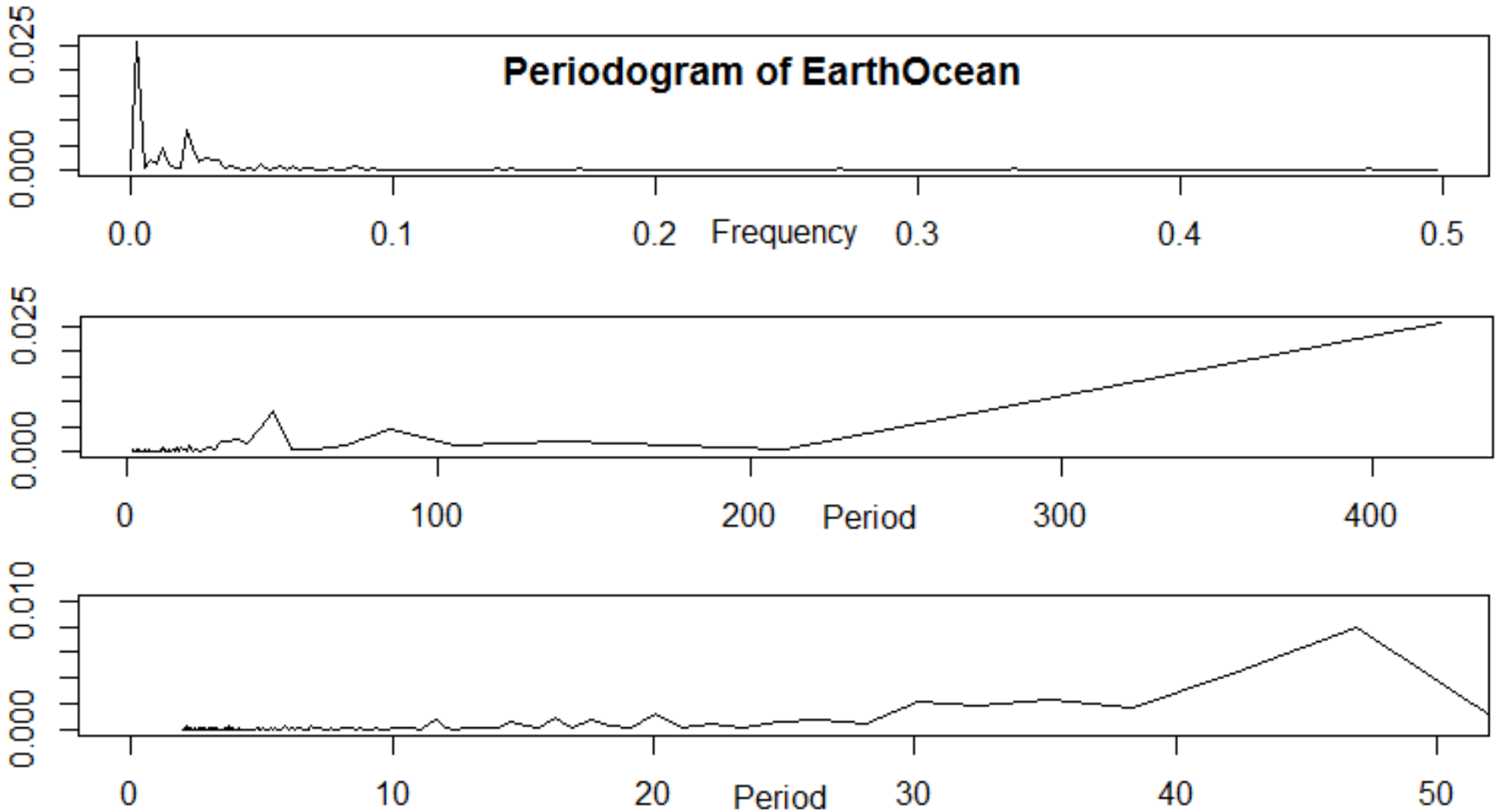


We can see an oscillation with a period ≈ 40 and the slow decay for small lags indicates the AR behavior.



Earth Ocean temperature: periodogram

® *pgram=Pgram(x,paste("Periodogram of",xname));*
dev.new()
plot(1/pgram[,1],pgram[,2], type="l", xlab="Period" ,
ylab="Scaled Periodogram")



Earth Ocean temp: estimation

® *arima(x,order=c(1,0,0)); arima(x,order=c(2,0,0)); arima(x,order=c(3,0,0)); arima(x,order=c(1,0,1)); arima(x,order=c(1,0,0),seasonal=list(order=c(1,0,0),period=44))*

$$x_t \overset{(1,0,0)}{=} 0.825x_{t-1} + \epsilon_t,\ \hat{\sigma}^2 = 0.0125$$

$$x_t \overset{(2,0,0)}{=} 0.57x_{t-1} + 0.31x_{t-2} + \epsilon_t,\ \hat{\sigma}^2 = 0.0113$$

$$x_t \overset{(3,0,0)}{=} 0.55x_{t-1} + 0.27x_{t-2} + 0.07x_{t-3} + \epsilon_t,\ \hat{\sigma}^2 = 0.0112$$

$$x_t \overset{(1,0,1)}{=} 0.93x_{t-1} + \epsilon_t - 0.36\epsilon_{t-1},\ \hat{\sigma}^2 = 0.0113$$

$$x_t \overset{(2,0,1)}{=} 0.78x_{t-1} + 0.14x_{t-2} + \epsilon_t - 0.22\epsilon_{t-1},\ \hat{\sigma}^2 = 0.0112$$

$$(1 + 0.37B)(1 - B)x_t \overset{(1,1,0)}{=} \epsilon_t,\ \hat{\sigma}^2 = 0.0118$$

$$(1 - 0.81B)(1 - 0.1B^{44})x_t \overset{(1,0,0)(1,0,0)}{=} \epsilon_t,\ \hat{\sigma}^2 = 0.0123$$

$$(1 - 0.57B - 0.3B^2)(1 - 0.1B^{44})x_t \overset{(2,0,0)(1,0,0)}{=} \epsilon_t,\ \hat{\sigma}^2 = 0.0112$$

Earth Ocean temp: Model identification

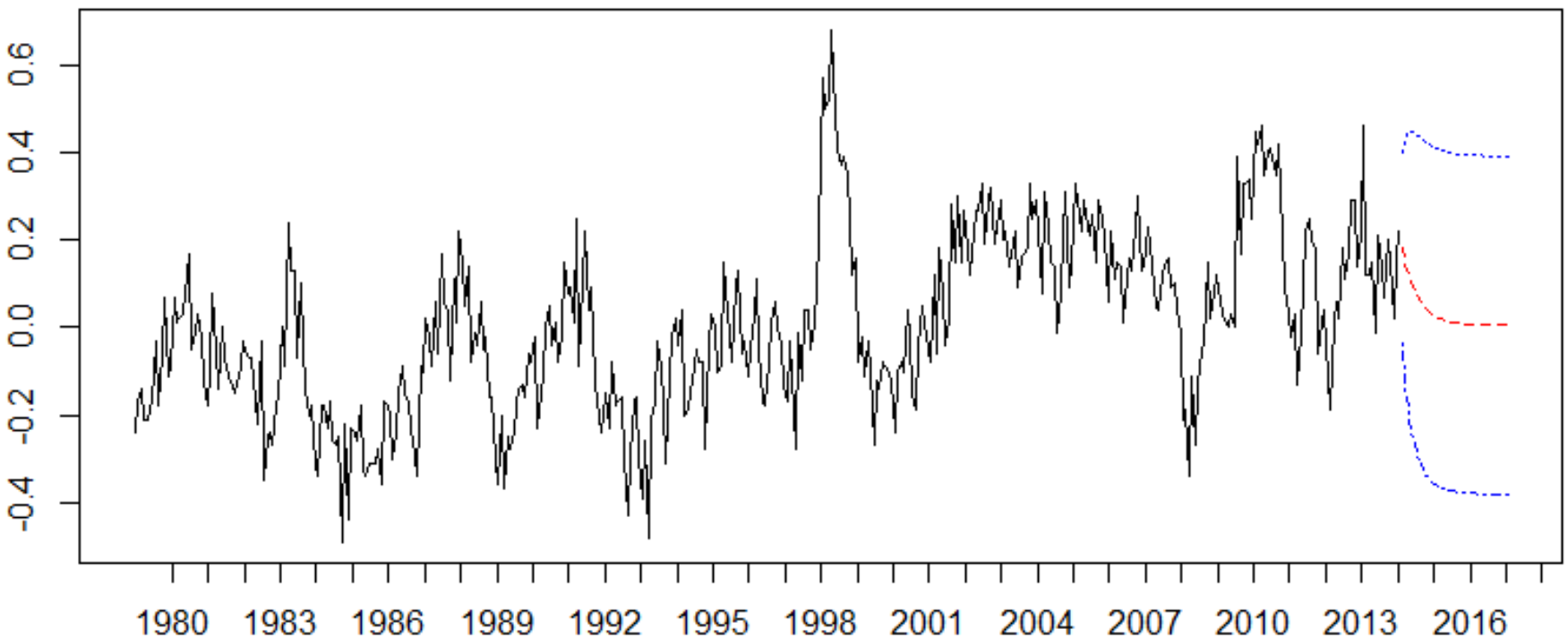
A good model has ...

- a small number of parameters, N (for easier interpretation)
- a small noise variance, $\hat{\sigma}^2$
- a large log-likelihood, LL
- a small Akaike Information Criterion, AIC

model	N	$\hat{\sigma}^2$	LL	AIC
ARMA(1,0)	1	0.01247	335.86	-665.73
ARMA(2,0)	2	0.01132	356.73	-705.45
ARMA(3,0)	3	0.01127	357.70	-705.40
ARIMA(1,1,0)	1	0.01182	347.12	-690.24
ARIMA(1,1,1)	2	0.01163	350.58	-695.17
ARIMA(1,0,1)	2	0.01129	357.30	-706.60
SARIMA(1,0,0)(1,0,0)^{44}	2	0.01237	337.45	-666.9

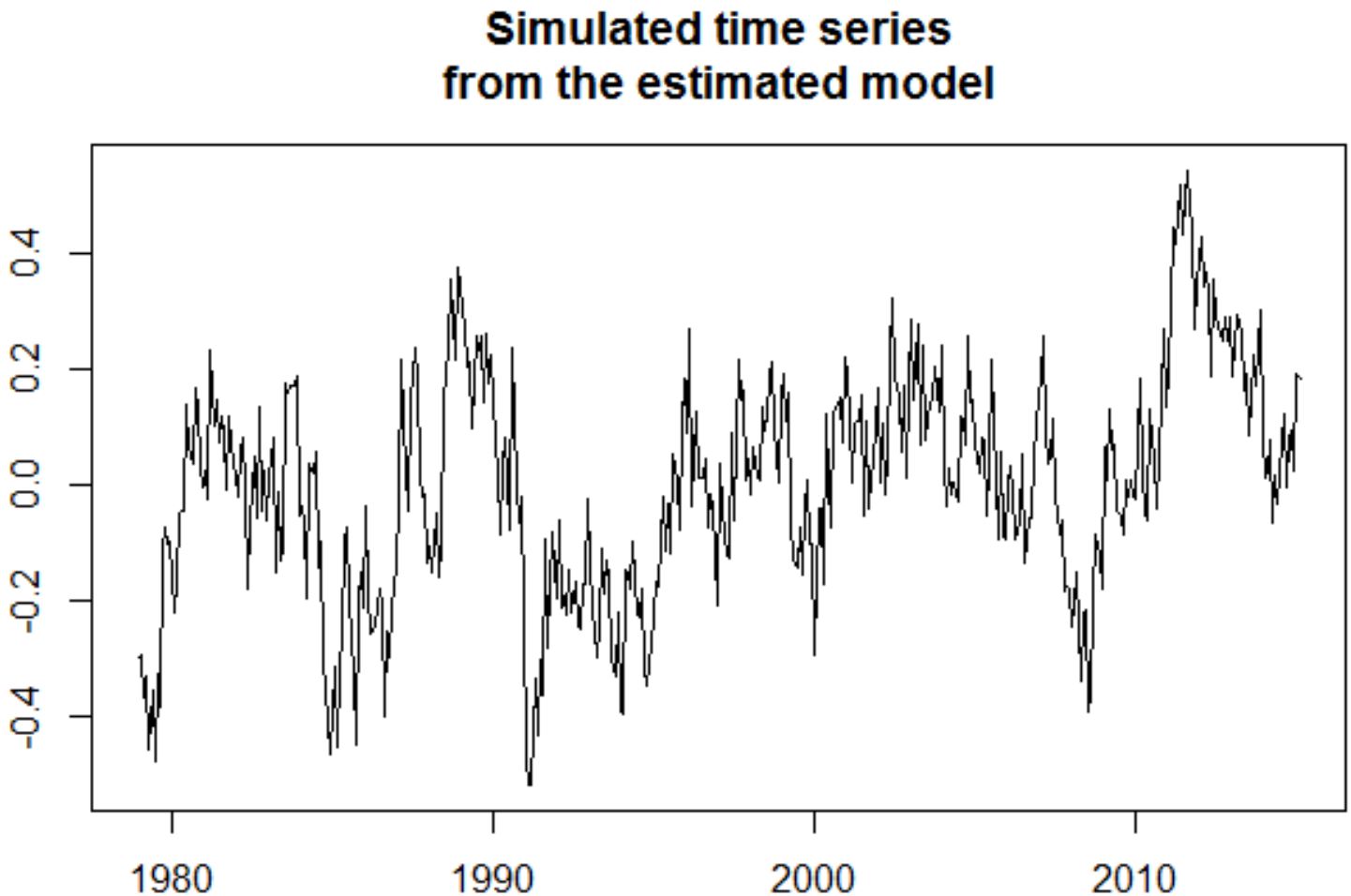
Earth Ocean temperature: forecasting

```
® h=36; model=arima(x,order=c(1,0,1))
pred=predict(model, n.ahead=h)$pred
s.pred=predict(model, n.ahead=h, se.t=T)$se
plot(t, x, type="l", xlim=c(t[1],t[n]+h/12),xaxt="n",main=xname)
lines(t[n]+(1:h)/12, pred, type="l", lty=2,col="red")
lines(t[n]+(1:h)/12, pred+1.96*s.pred, type="l", lty=3,col="blue")
lines(t[n]+(1:h)/12, pred-1.96*s.pred, type="l", lty=4,col="blue")
axis(1,at=c(1980:2020))
```



Earth Ocean temperature: simulation

```
ARmodel=arima(x,order=c(1,0,1))
simx=arima.sim(model=as.list(coef(ARmodel)),
sd=sqrt(ARmodel[["sigma2"]]), n=length(t))
plot(t, simx, type="l", xlim=c(t[1],t[n]),main="Simulated time series
from the estimated model")
```

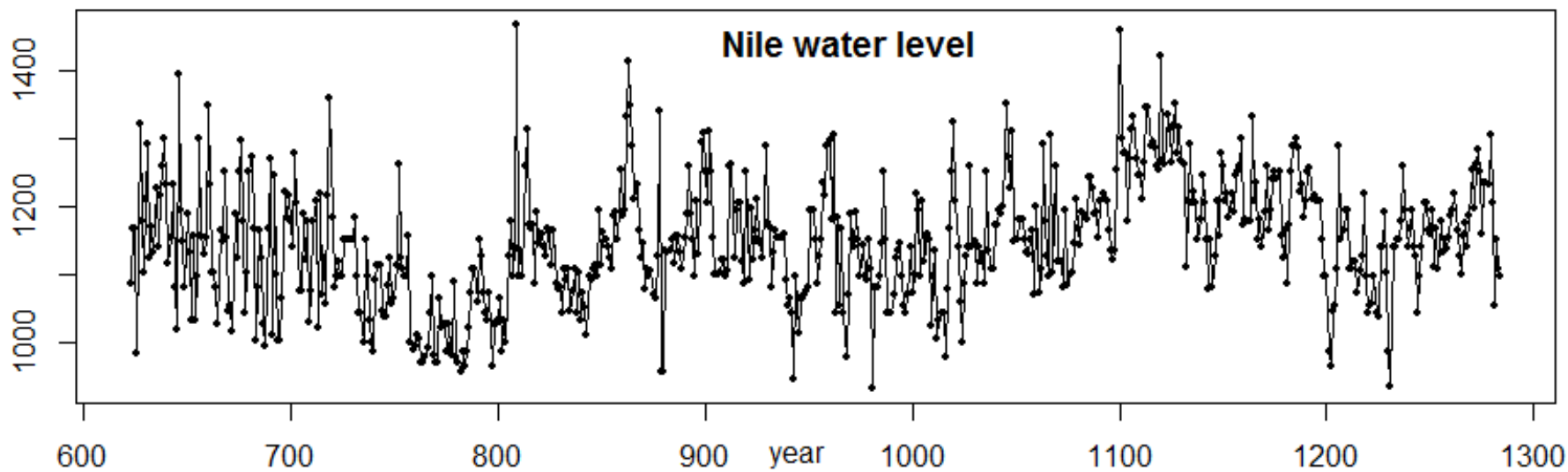


Nile water level

The time series is a record of the lowest annual water levels on the Nile river during 622-1284 measured at the island of Roda, near Cairo, Egypt.

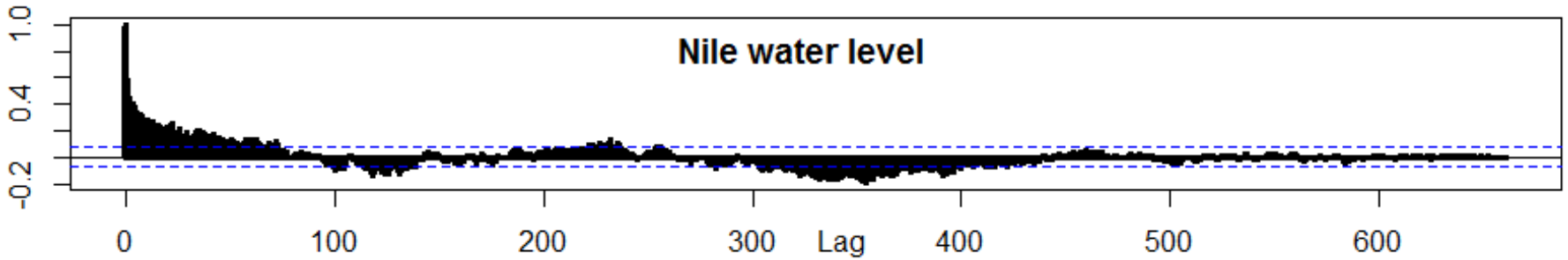
®

```
NileWLevel=read.table("http://ssa.cf.ac.uk/ma0367/nile-water-level.csv",
sep=",",header=TRUE)
print(NileWLevel); t=NileWLevel[,1]; n=length(t);
plot(t,NileWLevel[,2],type="l",main="Nile water level",xlab="year")
points(t,NileWLevel[,2],pch=19,cex=0.6)
```

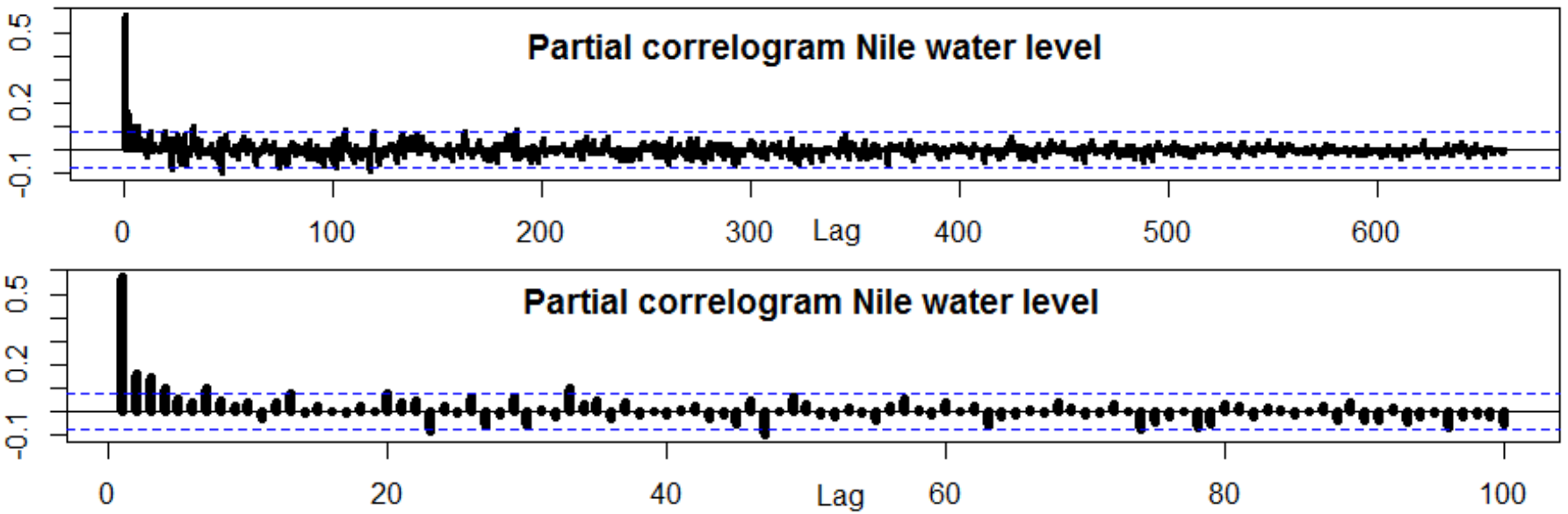


Nile water level: correlagram

® *corrgram=acf(NileWLevel[,2],n,lwd=4,main="Correlogram Nile water level")*



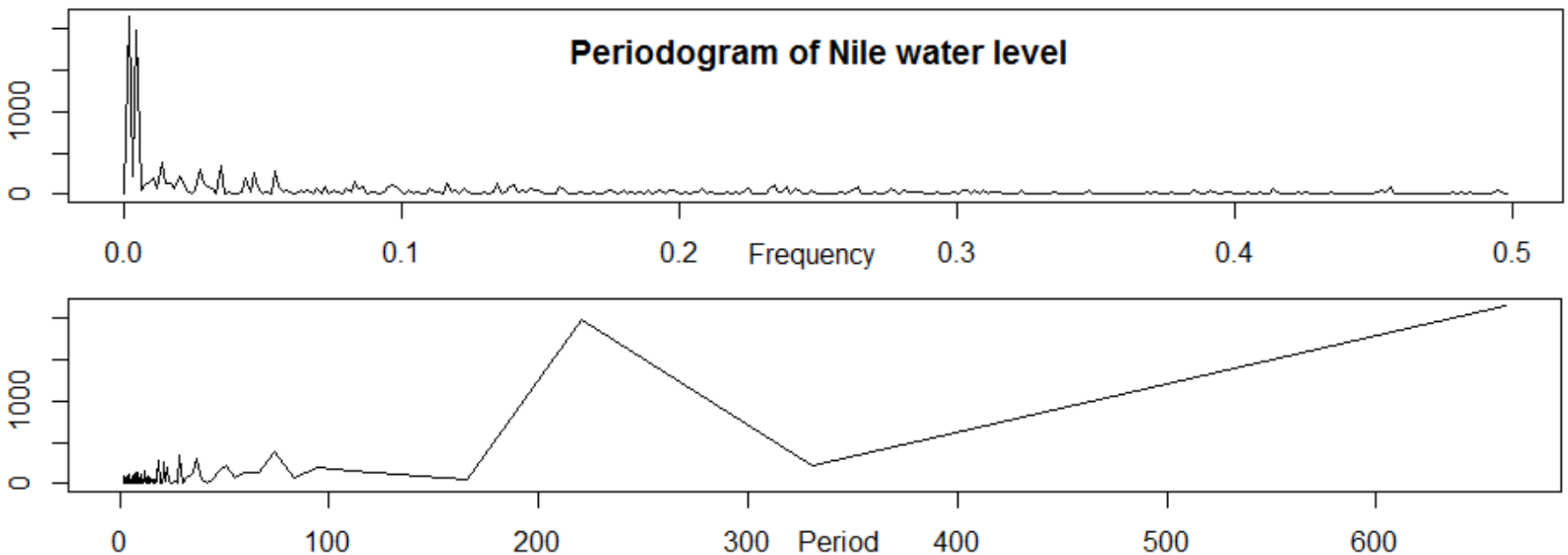
® *pacf(NileWLevel[,2],100,lwd=6,main="Partial correlogram Nile water level")*



The ARMA process should have at least 3 parameters.

Nile water level: periodogram

```
® mu=mean(NileWLevel[,2]);
pgram=Pgram(NileWLevel[,2],"Periodogram of Nile water level")
dev.new()
plot(1/pgram[,1],pgram[,2], type="l", xlab="Period" ,
ylab="Scaled Periodogram")
```



Nile water level: estimation

® *arima(NileWLevel[,2]-mu,order=c(1,0,0))*
arima(NileWLevel[,2]-mu,order=c(3,0,2))

$$x_t \overset{(1,0,0)}{=} 0.57x_{t-1} + \epsilon_t,\ \hat{\sigma}^2 = 5268.0$$

$$x_t \overset{(2,0,0)}{=} 0.48x_{t-1} + 0.16x_{t-2} + \epsilon_t,\ \hat{\sigma}^2 = 5136.0$$

$$x_t \overset{(3,0,0)}{=} 0.46x_{t-1} + 0.09x_{t-2} + 0.15x_{t-3} + \epsilon_t,\ \hat{\sigma}^2 = 5027.0$$

$$x_t \overset{(3,0,1)}{=} 1.28x_{t-1} - 0.3x_{t-2} - 0.01x_{t-3} + \epsilon_t - 0.86\epsilon_{t-1},\ \hat{\sigma}^2 = 4884.0$$

$$x_t \overset{(3,0,2)}{=} 2.03x_{t-1} - 1.23x_{t-2} + 0.2x_{t-3} + \epsilon_t - 1.6\epsilon_{t-1} + 0.66\epsilon_{t-2},\ \hat{\sigma}^2 = 4876.0$$

$$x_t \overset{(2,0,1)}{=} 1.28x_{t-1} - 0.3x_{t-2} + \epsilon_t - 0.86\epsilon_{t-1},\ \hat{\sigma}^2 = 4884.0$$

Nile water level: Model identification

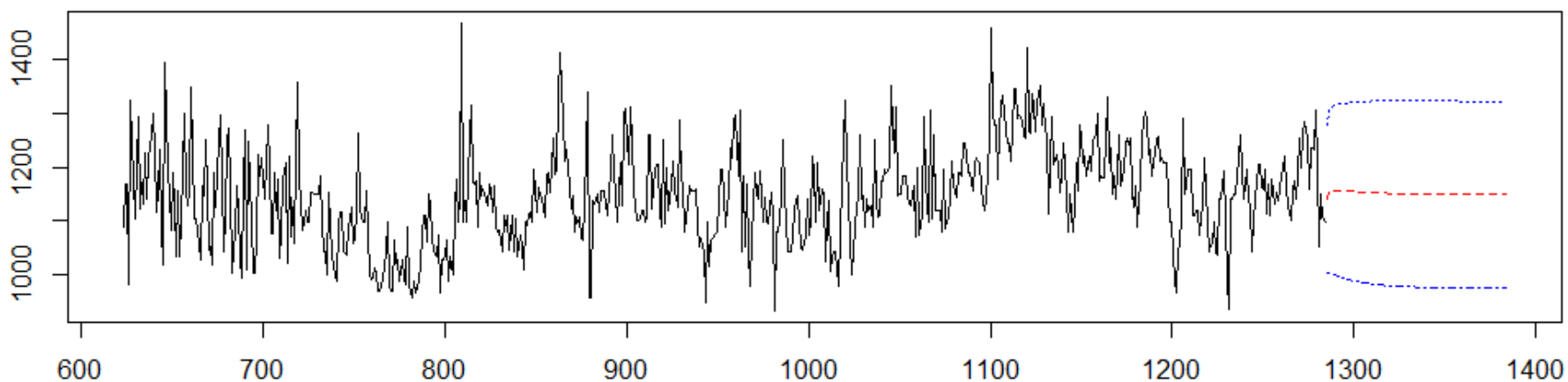
A good model has ...

- a small number of parameters, N (for easier interpretation)
- a small noise variance, $\hat{\sigma}^2$
- a large log-likelihood, LL
- a small Akaike Information Criterion, AIC

model	N	$\hat{\sigma}^2$	LL	AIC
ARMA(1,0)	1	5268	-3776.04	7558.07
ARMA(2,0)	2	5136	-3767.63	7543.27
ARMA(3,0)	3	5027	-3760.56	7531.13
ARMA(3,1)	4	4884	-3751.12	7514.25
ARMA(3,2)	5	4876	-3750.60	7515.21

Nile water level: forecasting

```
® h=100;
model=arima(NileWLevel[,2],order=c(2,0,1))
pred=predict(model, n.ahead=h)$pred
s.pred=predict(model, n.ahead=h, se.t=T)$se
plot(t, NileWLevel[,2], type="l", xlim=c(t[1],t[n]+h),xaxt="n")
lines(t[n]+(1:h), pred, type="l", lty=2,col="red")
lines(t[n]+(1:h), pred+1.96*s.pred, type="l", lty=3,col="blue")
lines(t[n]+(1:h), pred-1.96*s.pred, type="l", lty=4,col="blue")
axis(1,at=seq(600,1400,100))
```



ACF and PACF

Identifying the numbers of AR or MA terms in an ARIMA model
https://people.duke.edu/~rnau/411arim3.htm

Interpret the partial autocorrelation function (PACF)
https://support.minitab.com/en-us/minitab/18/help-and-how-to/modeling-statistics/time-series/how-to/partial-autocorrelation/interpret-the-results/partial-autocorrelation-function-pacf/

Learning with AI

Model Selection Diagnostics

Prompt: "I am comparing an ARMA(3,1) and an ARMA(2,0) for the Nile water level data. The ARMA(3,1) has a lower AIC but higher complexity. Explain the trade-off between Log-Likelihood and the number of parameters in the AIC formula."

Spectral Analysis and Periodograms

Prompt: "Looking at a periodogram (cpgram in R) for Earth Ocean temperature, I see a significant peak at frequency 0.025. Walk me through the math to convert this frequency into a seasonal period s for a SARIMA model."

Learning with AI, 2

Transformation Sensitivity

Prompt: "In the Airpass example, we used $y_t = \nabla \log x_t$ to achieve stationarity. Explain what might happen to the AIC and the variance of the residuals if I used a square-root transformation instead of a logarithm."

R Code Debugging

Task: Provide an AI with your R code for forecasting the Airpass data.

Prompt: "Explain why we use `cumsum(pred)` and `exp()` when reversing the transformations for a series that was originally log-differenced."

Homework

- Solve the past 2019 exam paper, which is provided separately on Learning Central.